

Non-Graded Optional Project: GenAI for Customer Segmentation and Insights Report

Project Title: Customer Segmentation and Insights Report

Objective

This project gives you hands-on experience with Generative AI for customer segmentation and insight analysis. By working through this project, you'll practice using AI to understand customer behavior, identify segments, and make data-driven recommendations—just like in the lessons.



Project Description

In this project, you'll use a Generative AI tool to analyze a customer dataset, identifying segments and generating insights to guide a fictional company's sales and marketing strategies. Your goal is to apply AI-driven segmentation techniques to create a report that highlights key customer segments, provides actionable insights, and suggests strategies to engage each segment effectively.

Steps for the Project

Step 1: Review the Customer Data

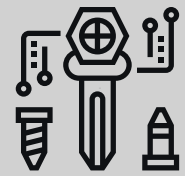


Open the Dataset: Access the dataset provided in the course or use your own dataset. The dataset includes customer information such as age, location, purchase history, and engagement metrics.



Understand the Data Structure: Familiarize yourself with the types of information available. Make sure to take note of key columns like purchase frequency, age, and engagement level, as these will be useful in AI-driven segmentation.

Step 2: Use GenAI to Segment Customers



Choose a GenAI Tool: Use ChatGPT or a similar AI tool capable of data analysis and segmentation.



Prompt the AI for Segmentation: Enter prompts to segment the customers based on behaviors, demographics, and engagement levels. Here are a few prompt examples you can use based on the techniques covered in the lessons:

"Segment the customer data into categories such as high spenders, frequent buyers, and low-engagement customers. Provide descriptions for each segment."



"Analyze the dataset and identify distinct customer groups based on purchase history and engagement. Summarize each group's common characteristics."

"Identify key customer segments based on demographics and behaviors, and provide suggestions for engaging each segment."



Record AI-Generated Segments: Note down the segments identified by the AI. Example segments might include "High Spenders," "New Customers," "Occasional Buyers," and "Engaged Shoppers."

Step 3: Analyze AI Insights for Each Segment

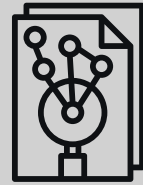


Segment Analysis Using AI: Use AI to analyze each segment's demographics, behavior patterns, and engagement levels. You might ask questions such as:

"What are the main behaviors of high spenders in this dataset?"

"What patterns can we observe for customers with high engagement?"

"Provide a summary of key demographic trends within the frequent buyer segment."



Document AI-Generated Insights: Record the AI's findings for each segment. These insights could include factors such as average purchase amount, product preferences, and common engagement behaviors.

Step 4: Identify Opportunities and Challenges for Each Segment



Prompt AI for Opportunities and Challenges: Use AI to identify specific opportunities and challenges associated with each segment.
For example:



"List potential opportunities for increasing engagement with low-engagement customers."

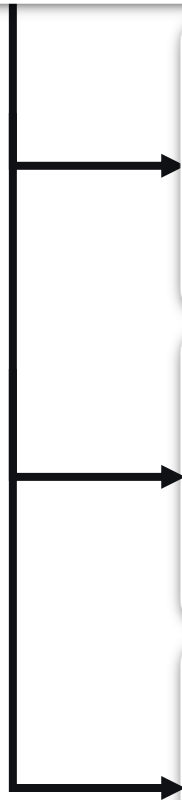
"What challenges might we face in retaining high-spending customers?"



Record Responses: Summarize the AI-generated opportunities and challenges, such as potential for upselling high spenders or engaging new customers with onboarding materials.

Step 5: Develop AI-Driven Recommendations for Each Segment

Based on your AI-powered segmentation, create actionable recommendations for each segment to improve engagement and increase sales. Use AI to refine your recommendations. Suggested prompts include:



"Provide a targeted marketing strategy for frequent buyers based on their behavior and preferences."

"Suggest personalized messages for low-engagement customers to increase their activity."

"What types of promotions might work best for new customers to encourage repeat purchases?"



Finalize Recommendations: For each segment, provide at least one specific recommendation, such as offering loyalty rewards for high spenders or sending welcome discounts to new customers.

Step 6: Compile the Report

Organize your findings in a concise report format:

→ **Title Page:** Project title, your name, and date.

→ **Customer Segmentation Overview:** Briefly summarize the segments identified by the AI.

→ **Segment Analysis:** Describe the demographics, behaviors, and engagement trends for each segment, referencing AI-generated insights.

→ **Opportunities and Challenges:** Highlight one opportunity and one challenge for each segment, as identified by AI.

→ **Recommendations:** Provide AI-driven recommendations for each segment.

Step 7: Share the Project

Save your report in PDF format.

Upload your report to the peer review area or share it with peers for feedback.

Instructions for Documenting and Sharing The Project for Peer Review

1. Document the Project

Use word processing software to create your report.

Ensure all sections of the project document structure are completed.

Proofread and edit your report for clarity and accuracy.

2. Share the Project

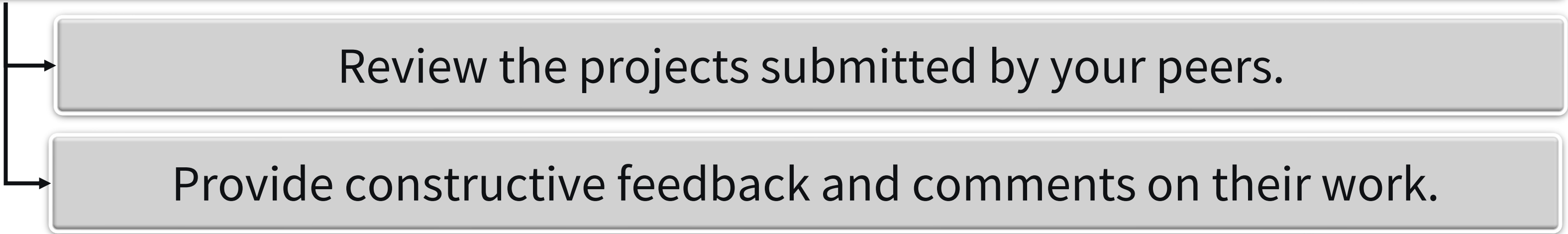
Save your report in PDF format.

Upload your documents to the “Peer Review” area in the course shell.

Provide a brief description of your project in the submission post.

Instructions for Documenting and Sharing The Project for Peer Review

3. Peer Review



Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.

Skills Developed:



Practical application of AI for customer segmentation.



Interpretation of AI-driven insights to inform sales strategies.



Creation of targeted recommendations to optimize customer engagement and sales.

This project will help you solidify your understanding of using AI in sales analytics and prepare you to make data-driven decisions for real-world sales scenarios.